

Week 7, Lecture 2: Bayesian Inference

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** So far we have treated a parameter as a fixed but unknown number and described our uncertainty with confidence intervals. Bayesian inference describes that uncertainty with a probability distribution over the parameter values. Today we build that idea from Bayes' rule, work one example all the way through, and talk about why Bayesian methods have become so common.
- **Read:** Bolker 6.2.2 (Bayesian analysis, printed p. 237-243) and 6.4.2 (posterior and marginal distributions, p. 258-260).
- **Review:** Bayes' rule from Week 3 (Bolker 4.3, p. 145-152, especially 4.3.3 on priors), and the Beta distribution (Bolker 4.5.2.5, p. 176). Look back at the Week 5 WOD (*Writing the Likelihood*) and your likelihood values from Part 2.
- **Wednesday's take-home WOD** will be handed out at the start of class and is due Monday.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

Beat	Time
Hand out take-home WOD (<i>Curvature by Hand</i>)	~3 min
Lecture	~37 min
Discussion	~8 min

No in-class WOD today. Hand out `wod/WOD_Week7_Lecture2_CurvatureByHand.Rmd`, say it's due Monday at the start of class, and point out that Part 3 is the only part that needs R. Budget time Monday to go over it (key in `wod/answer_keys/`).

Source material: Bolker 4.3.3 (priors, and why flat priors aren't neutral), 6.2.2.1 (binomial/Beta conjugate example; the numbers below are his, recast as a germination trial), 6.4.2 (credible intervals); legacy `LECTURE6.Rmd` ("Bayesian Analysis #1: Concepts": prior $\times$ likelihood figures, brute-force Bayes, mean vs. mode, credible intervals); archival `LAB4.Rmd` for the Stan preview (not used today).

Example choice. The conjugate example is a seed-germination trial rather than Bolker’s tadpoles. The numbers (30 of 40, and a prior worth 120 of 200) are Bolker’s from p. 237-238, so students who did the reading can check them, but the setting is one the agricultural and restoration students will recognize. The opening grid example reuses the Week 5 WOD nest data, because students already computed those likelihoods by hand.

Where this sits in the week

- **Monday:** uncertainty from the shape of the likelihood: likelihood ratio intervals, profiles, the Hessian. Section 3 of Monday’s lecture gave a four-minute preview of today (prior $\times$ likelihood on the pika data, credible interval [1.91, 4.74]).
- **Tuesday (Lab 6):** Bliss’s beetle fumigation data. Profile and Hessian intervals for the LD50.
- **Today:** Bayesian inference from the ground up. Keep the math to one-parameter examples; MCMC and Stan come later.
- **Coming up:** optimization (Week 8), then MCMC and Lab 7 on Bayesian inference (Week 9).

Part 1: Lecture content

1. Big picture (~3 min)

Loop on the board:

THEORY $\rightarrow$ MODEL(S) $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat

- The model and the likelihood don’t change today. What changes is how we turn the likelihood into a statement about the parameter.
- **Frequentist view (Monday):** the parameter is a fixed, unknown number. The data are random. A 95% confidence interval is a procedure that captures the true value in 95% of repeated experiments.
- **Bayesian view (today):** our uncertainty about the parameter is itself described by a probability distribution over the parameter values. Before the data, that distribution is the **prior**; after, it’s the **posterior**. The likelihood is what connects them.
- The tool that does the updating is one students already know: Bayes’ rule, from Week 3 and Lab 3’s camera-trap classifier.

2. Bayes’ rule, from a few hypotheses to a whole distribution (~7 min)

Recall Week 3:

$$P(H_i | D) = \frac{P(D | H_i) P(H_i)}{\sum_j P(D | H_j) P(H_j)}.$$

In Lab 3, the hypotheses were “target species” and “not.” Now make the hypotheses be values of a parameter.

Step 1: three hypotheses. Week 5 WOD nest data: 4 nests, outcomes 1, 1, 0, 1, so $\mathcal{L}_4(p) = p^3(1 - p)$. Treat $p = 0.25, 0.5, 0.75$ as three competing hypotheses, equally likely beforehand.

```
p3 <- c(0.25, 0.5, 0.75)
lik3 <- p3^3 * (1 - p3)
prior3 <- rep(1/3, 3)
post3 <- lik3 * prior3 / sum(lik3 * prior3)
round(rbind(likelihood = lik3, posterior = post3), 3)
```

```
##           [,1] [,2] [,3]
## likelihood 0.012 0.062 0.105
## posterior  0.065 0.348 0.587
```

```
sum(lik3)           # Week 5 WOD 2d: likelihoods don't sum to 1
```

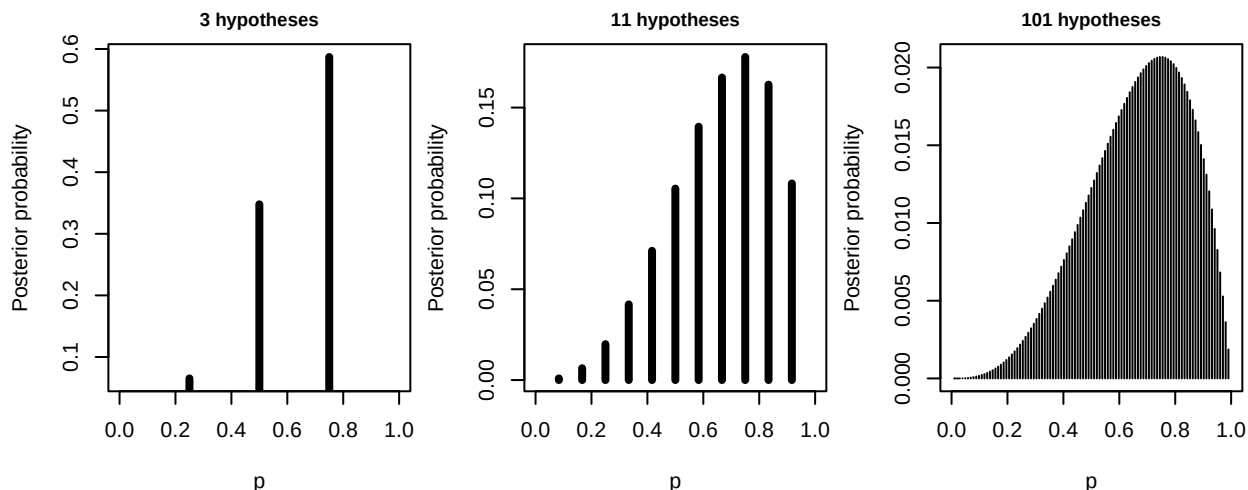
```
## [1] 0.1796875
```

- Students computed these likelihoods in the Week 5 WOD (3/256, 16/256, 27/256) and found they sum to 46/256, not 1. Bayes’ rule fixes that: multiply by the prior, divide by the total. Posterior probabilities 3/46, 16/46, 27/46.
- With equal priors, the posterior is just the likelihood rescaled to sum to 1 (Bolker eq. 4.3.16).

Step 2: more hypotheses. Nothing stops us using 100 or 10,000 values of p . The sum becomes an integral and the prior and posterior become probability **densities**:

$$\underbrace{P(p | y)}_{\text{posterior}} = \frac{\overbrace{\mathcal{L}(p)}^{\text{likelihood}} \overbrace{P(p)}^{\text{prior}}}{\int_0^1 \mathcal{L}(p) P(p) dp}$$

```
par(mfrow = c(1, 3), mar = c(4, 4, 2, 0.5))
for (n_grid in c(3, 11, 101)) {
  p_g <- seq(0, 1, length.out = n_grid + 2)[-c(1, n_grid + 2)]
  w <- p_g^3 * (1 - p_g)
  plot(p_g, w / sum(w), type = "h", lwd = if (n_grid < 20) 4 else 1,
       xlim = c(0, 1), xlab = "p", ylab = "Posterior probability",
       main = paste(n_grid, "hypotheses"), cex.main = 0.9)
}
```



- The denominator is a single number, the **marginal likelihood**: the probability of the data averaged over all parameter values. It only rescales the curve to area 1. For one parameter it's easy (a sum or an integral). For 20 parameters it's a 20-dimensional integral, which is why MCMC exists (later in the semester).
- Notation: the posterior $P(p \mid y)$ is a real conditional probability, so it gets a bar. The likelihood $\mathcal{L}(p)$ still doesn't.

Sticking point: students conflate “a probability distribution for p ” with “ p is random, like a coin flip.” Say plainly: to a Bayesian the posterior describes *our knowledge* of p , not variation in p itself. The germination rate is one number; we're uncertain about it.

3. A worked example: the Beta prior (~8 min)

Setting. A restoration crew tests a native seed lot: 30 of 40 seeds germinate. What is the lot's germination probability p ?

The likelihood is binomial: $\mathcal{L}(p) \propto p^{30}(1-p)^{10}$.

The prior. A Beta(α, β) density is $P(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$. Same shape as a binomial likelihood. Multiply them:

$$P(p \mid y) \propto p^{30+\alpha-1}(1-p)^{10+\beta-1},$$

which is another Beta: **Beta**($\alpha + k, \beta + n - k$). A prior that gives a posterior from the same family is called **conjugate**. It means no integral: just add counts.

- Interpretation (Bolker p. 237): $\alpha - 1$ and $\beta - 1$ act like earlier successes and failures. Beta(1, 1) is flat: no prior seeds.
- **Flat prior:** posterior Beta(31, 11).
- **Informative prior:** the supplier's records say 120 of 200 seeds from earlier lots germinated. Prior Beta(121, 81); posterior Beta(151, 91).

```

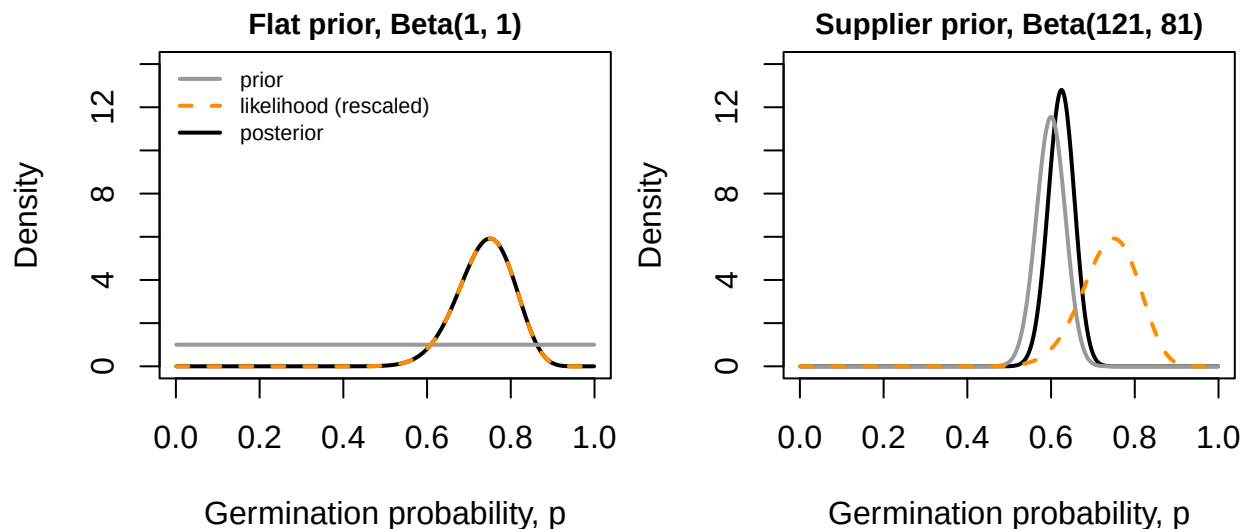
k <- 30; n <- 40
post_flat <- c(1 + k, 1 + n - k)
post_info <- c(121 + k, 81 + n - k)
beta_mean <- function(ab) ab[1] / sum(ab)
beta_mode <- function(ab) (ab[1] - 1) / (sum(ab) - 2)
summ <- rbind(
  flat = c(mean = beta_mean(post_flat), mode = beta_mode(post_flat),
    qbeta(c(0.025, 0.975), post_flat[1], post_flat[2])),
  informative = c(mean = beta_mean(post_info), mode = beta_mode(post_info),
    qbeta(c(0.025, 0.975), post_info[1], post_info[2]))
)
colnames(summ)[3:4] <- c("lower95", "upper95")
round(summ, 3)

##           mean  mode lower95 upper95
## flat       0.738 0.750   0.597   0.858
## informative 0.624 0.625   0.562   0.684

f2 <- function(v) formatC(v, format = "f", digits = 2)

p_seq <- seq(0.001, 0.999, length.out = 500)
lik_scaled <- dbeta(p_seq, k + 1, n - k + 1) # likelihood rescaled to area 1
plot_panel <- function(prior_ab, post_ab, title) {
  plot(p_seq, dbeta(p_seq, post_ab[1], post_ab[2]), type = "l", lwd = 2,
    ylim = c(0, 14), xlab = "Germination probability, p", ylab = "Density",
    main = title, cex.main = 0.9)
  lines(p_seq, dbeta(p_seq, prior_ab[1], prior_ab[2]), lwd = 2, col = "grey60")
  lines(p_seq, lik_scaled, lwd = 2, lty = 2, col = "darkorange")
}
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
plot_panel(c(1, 1), post_flat, "Flat prior, Beta(1, 1)")
legend("topleft", c("prior", "likelihood (rescaled)", "posterior"),
  lwd = 2, lty = c(1, 2, 1), col = c("grey60", "darkorange", "black"),
  bty = "n", cex = 0.7)
plot_panel(c(121, 81), post_info, "Supplier prior, Beta(121, 81)")

```



- **Flat prior:** the posterior *is* the rescaled likelihood. Its mode is 0.75, the MLE; its mean is 0.738, slightly pulled toward 0.5.
- **Supplier prior:** the posterior sits between prior and data, closer to the prior (mean 0.624), because the prior is “worth” 200 seeds and the data only 40.
- **More data wins.** If the crew had tested 400 seeds with 300 germinating, the same supplier prior gives a posterior mean of 0.699: the data now outweigh the prior two to one.
- **Updating in sequence.** Today’s posterior is tomorrow’s prior. Testing 40 seeds and then 40 more gives the same answer as testing 80 at once (Bolker p. 237).

4. Summarizing a posterior (~4 min)

- **Point estimates.** Posterior mean (most common), median, or mode. With a flat prior, the mode equals the MLE. For skewed posteriors the three differ; Bolker prefers the mean because it uses the whole distribution.
- **Credible interval.** The central 95% of the posterior (2.5% and 97.5% quantiles), or the highest posterior density (HPD) interval, the shortest interval holding 95% (Bolker p. 258-259). For symmetric posteriors they coincide.
- **Interpretation.** “Given the model, the prior, and the data, there is a 95% probability that p is between 0.60 and 0.86.” That is the sentence most people *want* to say about a confidence interval (Monday’s discussion question 1), and a Bayesian is allowed to say it.

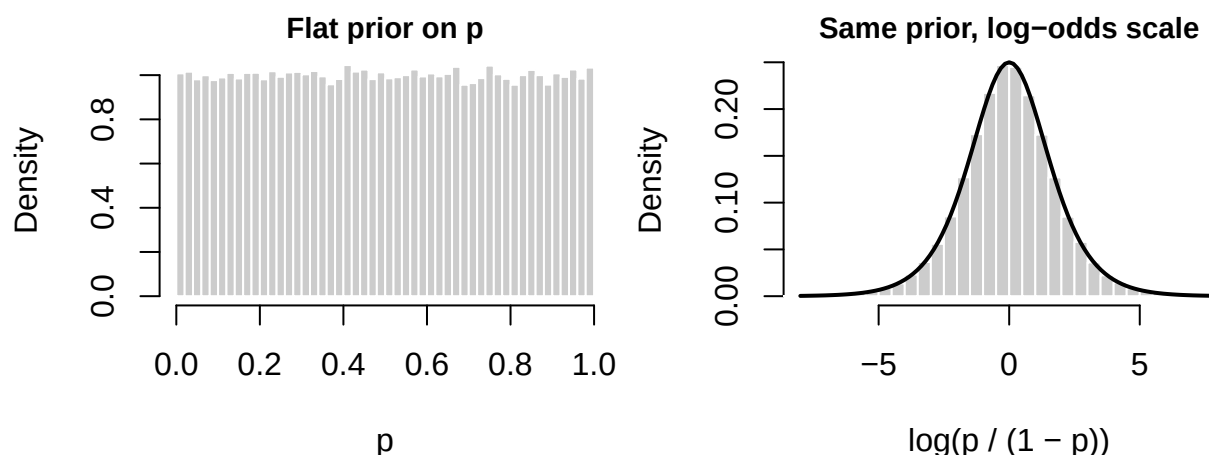
5. Where new Bayesians get stuck (~8 min)

Pick four or five; each takes a minute or two.

(a) “**The likelihood is the posterior.**” Only after multiplying by a prior and rescaling to area 1. The likelihood alone doesn’t integrate to 1 (Week 5 WOD) and isn’t a distribution for p .

(b) **“A flat prior means no information.”** Flat on one scale isn’t flat on another (Bolker 4.3.3, the bird-mass example). A uniform prior on p implies a prior on the log-odds, $\log \frac{p}{1-p}$, that is peaked at 0:

```
set.seed(746)
p_draws <- runif(1e5)
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
hist(p_draws, breaks = 40, freq = FALSE, col = "grey80", border = "white",
     main = "Flat prior on p", xlab = "p", cex.main = 0.9)
hist(qlogis(p_draws), breaks = 60, freq = FALSE, col = "grey80", border = "white",
     main = "Same prior, log-odds scale", xlab = "log(p / (1 - p))",
     xlim = c(-8, 8), cex.main = 0.9)
curve(dlogis(x), add = TRUE, lwd = 2)
```



So “uninformative” always means “uninformative on some scale.” With plenty of data it rarely matters; with little data it can.

(c) **“Priors let you get any answer you want.”** In principle yes (Bolker p. 150: a prior of 0 on a hypothesis means no data can ever support it). In practice: priors must be reported, reviewers can object, and the standard move is a **prior sensitivity analysis**: rerun with a few reasonable priors and see whether the conclusion changes. If it does, the data are weak and that’s worth knowing.

(d) **“Bayesian means MCMC.”** MCMC is a computing method for approximating posteriors when the denominator can’t be computed. The germination example needed none. Conversely, MCMC can be used for non-Bayesian problems too.

(e) **“The credible interval and the confidence interval are the same thing.”** With a flat prior and plenty of data the numbers are often close (Monday: likelihood-ratio [1.82, 4.61] vs. credible [1.91, 4.74] for the pikas), but the statements differ: a credible interval is a probability statement about the parameter; a confidence interval is a statement about the procedure.

(f) **“A high posterior probability proves the model.”** Every posterior is conditional on the model being right. A Bayesian model still has to be confronted with data (does it predict

data that look like what we saw?) and compared with rivals. The loop on the board doesn't change.

6. Why Bayesian methods are so widely used (~5 min)

Interpretation.

- Direct probability statements about parameters and hypotheses: “95% probability the germination rate is between 0.60 and 0.86.”
- Natural accumulation of evidence: last year's posterior is this year's prior.

Practical.

- **Uncertainty in derived quantities is easy to get.** Callback to Lab 6: for the LD50, a Bayesian draws many (a, b) pairs from the posterior and computes $-a/b$ for each; the spread of those values is the posterior for the LD50. No profile, no reparameterization.
- **Hierarchical models.** Plots within sites within regions, animals measured repeatedly, many species sharing information: these models have many parameters, some describing variation among groups. They are awkward to fit by maximum likelihood and natural in a Bayesian framework (later in the semester).
- **Hidden quantities.** True occupancy when detection is imperfect (the frog from Lab 5), true infection status behind a noisy test, the true population size behind a count. These are just more unknowns in a Bayesian model.
- **Small samples.** No reliance on large-sample approximations like the Hessian interval that went negative for the chytrid data (the take-home WOD shows this).
- **Prior information.** Previous studies, expert knowledge, or physical limits can be built in openly.

Costs. Priors must be chosen and defended; computation can be slow; MCMC output has to be checked for convergence.

7. Return to the big picture (~1 min)

- Same model, same likelihood. Bayesian inference multiplies by a prior and reads uncertainty off a posterior distribution.
- With one parameter and a conjugate prior, it's just adding counts. With many parameters, we need MCMC (Week 9).

Discussion (~8 min)

Pick one or two:

1. A restoration manager says, “I don't want the supplier's claims anywhere near my germination estimate.” Is that a reasonable position? What would you show them?
2. Two analysts fit the same germination data with “flat” priors, one flat on p and one flat on the log-odds of p . Could they get different answers? When would it matter?

3. In Lab 6 you built a profile interval for the LD50. How would a Bayesian get an interval for the LD50, and which approach would you rather use for a model with ten parameters?

Question 1 surfaces the “priors are cheating” worry. Question 3 connects today to Tuesday’s lab and sets up MCMC.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

Take-home WOD ~3 min. Lecture ~37 min. Discussion ~8 min.

- Hand out *Curvature by Hand*; due Monday; only Part 3 needs R

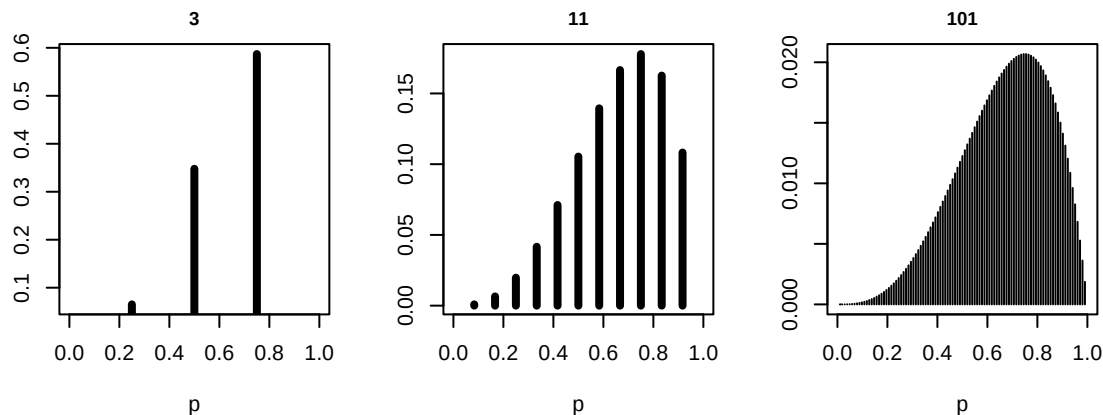
1. Big picture (~3 min)

- Loop on board: THEORY $\rightarrow$ MODEL $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat
- Same likelihood; new way to describe uncertainty
- Frequentist: parameter fixed, data random, CI = procedure
- Bayesian: uncertainty about parameter = probability distribution
- Prior $\rightarrow$ (likelihood) $\rightarrow$ posterior, via Bayes' rule (Week 3, Lab 3)

2. Bayes' rule, few hypotheses to many (~7 min)

$$P(H_i | D) = \frac{P(D | H_i)P(H_i)}{\sum_j P(D | H_j)P(H_j)}$$

- Week 5 nests: $\mathcal{L}_4(p) = p^3(1 - p)$
- $p = 0.25, 0.5, 0.75$: likelihoods 3, 16, 27 (/256), sum 46/256
- Equal priors $\rightarrow$ posterior 3/46, 16/46, 27/46



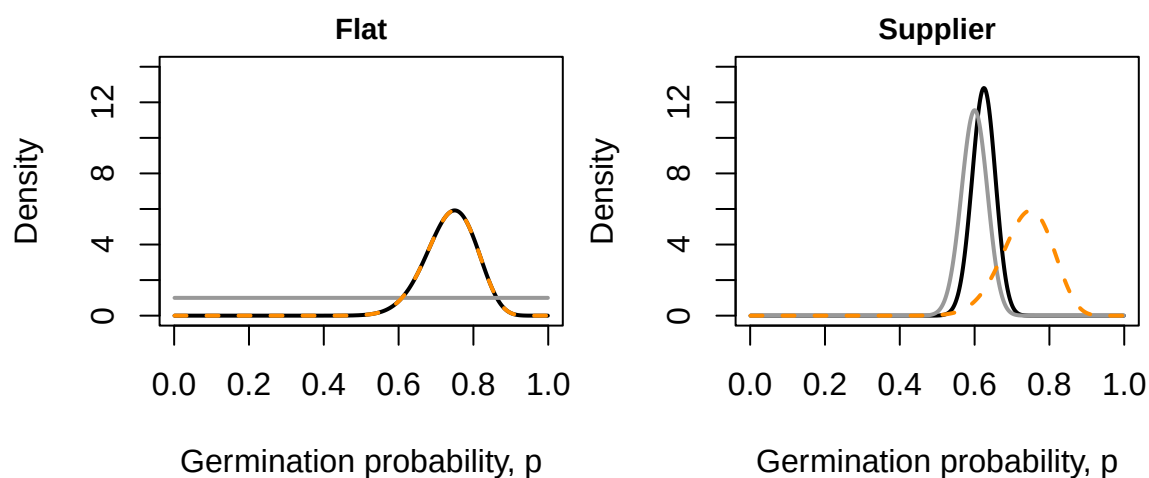
$$P(p | y) = \frac{\mathcal{L}(p) P(p)}{\int \mathcal{L}(p) P(p) dp}$$

- Denominator = one number; hard in many dimensions $\rightarrow$ MCMC
- **Sticking point:** posterior = our knowledge of p , not p varying

3. Beta prior (~8 min)

Seeds: **30 of 40** germinate. Supplier: 120 of 200.

- Beta prior $\times$ binomial $\rightarrow$ Beta($\alpha + k$, $\beta + n - k$)
- Conjugate = add counts; $\alpha - 1$, $\beta - 1$ = prior successes, failures
- Flat: Beta(31, 11); mean 0.738, mode 0.75 = MLE
- Supplier: Beta(151, 91); mean 0.624
- 300 of 400 with supplier prior: mean 0.699
- Sequential updating = same answer

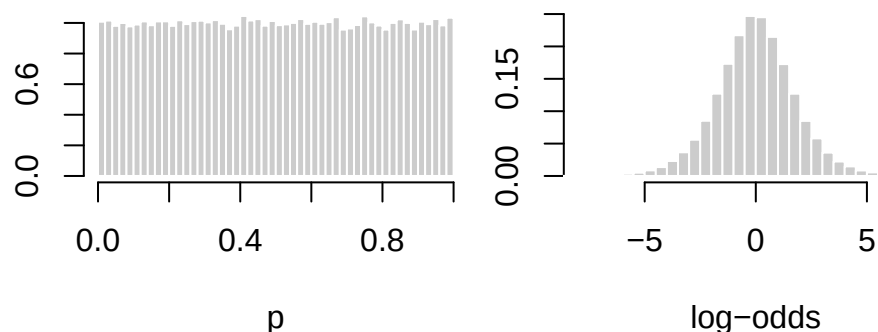


4. Summaries (~4 min)

- Mean (usual), median, mode (= MLE if flat)
- Credible interval: quantiles or HPD
- Flat: [0.60, 0.86]; supplier: [0.56, 0.68]
- “95% probability p is in here” is allowed

5. Where people get stuck (~8 min)

- (a) Likelihood $\neq$ posterior (Week 5 WOD)
- (b) Flat isn't flat on every scale



- (c) Priors: report them; sensitivity analysis
- (d) Bayes $\neq$ MCMC
- (e) Credible $\neq$ confidence (pikas: [1.82, 4.61] vs. [1.91, 4.74])
- (f) Posterior is conditional on the model; still check and compare

6. Why so popular (~5 min)

- Direct probability statements
- Evidence accumulates: posterior $\rightarrow$ next prior
- Derived quantities: $LD50 = -a/b$ for each posterior draw (Lab 6)
- Hierarchical models (later)
- Hidden quantities: occupancy, true infection status
- Small samples: no Hessian approximation (chytrid)
- Costs: priors, computation, convergence

7. Wrap (~1 min)

- Conjugate = add counts; many parameters = MCMC (Week 9)

Discussion (~8 min)

1. Manager: “keep supplier claims out of my estimate.” Reasonable?
2. Flat on p vs. flat on log-odds: different answers?
3. Bayesian interval for the LD50? Which approach for 10 parameters?